Environmental Pollution

Our planet and its environment are pure, clean and healthy for all living beings. But the increased human population puts pressure on the natural resources of the environment. The activities and lifestyles of people are changing with the evolution of human civilization. The trend of industrialization and urbanization is increasing. This leads to contamination of air, water, soil and everything in the earth which is called pollution that makes the nature dead.

Any undesirable change in physical, chemical or biological characteristics of air, water and soil which harmfully affect the life of humans and other desired species is called pollution. Pollution leads our planet on a path of destruction and has become a major issue across the globe.

Types of Pollutants

i. Primarily Pollutants

The pollutants that directly cause pollution after emitted from the source are known as primary pollutants. *e.g.,* sulfur-dioxide emitted from factories.

ii. Secondary Pollutants

These are formed by the intermingling and reactions of primary pollutants. *e.g.,* smog produced by the interactions of several primary pollutants.

iii. Biodegradable Pollutants

These pollutants are rapidly decomposed under natural condition by number of microorganisms. However, decomposition, becomes difficult when their rate of accumulation exceeds the rate of decomposition. *e.g.,* dometstic sewage, cloth, paper, woods, etc.

iv. Non-degradable Pollutants

These pollutants are neither decomposed by microorganisms nor broken down into pieces by physical or chemical agents of environment. *e.g.,* aluminium cans, mercuric salts, DDT, glass, etc. They are accumulated in the food chain and are biologically magnified.

Biodegradable Pollutants	Non-biodegradable Pollutants
1. They are decomposed and degraded by microbes into harmless materials.	1. They are not degraded or broken by microbes.
2. They are recycled naturally or artificially.	2. They are not recycled and cause pollution.
3. They are made up of natural ingredients. Examples: Cow dung, paper, etc.	3. They are mostly synthetic ingredients. Examples: DDT, polyethylene, etc.

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Primary Air Pollutants	Secondary Air Pollutants
1. These pollutants enter the environment directly from the source.	1. They develop from interaction of primary pollutants.
2. Photochemical reactions do not occur.	2. These are formed through photochemical reactions.
3. No enhanced effect. e.g., SO_2, CO, etc.	3. They are more potent due to synergism e.g., H_2SO_4, O_3, Nitrogen oxides, etc.

Types of Pollution: Pollution occurs in many forms, including the following.

1. Air pollution
2. Water pollution
3. Soil pollution
4. Marine pollution
5. Noise pollution
6. Radioactive pollution
7. Light pollution

Air Pollution: *The Silent Killer*

Any undesirable change to the normal composition of the air which harmfully affects the humans and other desirable species is called air pollution.

Sources of Air Pollution: The major outdoor pollution sources include the following.

i. Exhaust from factories and industries

The burning of fossil fuels like coal and mineral oils and release of wastes in manufacturing industries, petrochemical industries, power plants, iron and steel industries, fertilizer industries, cotton industries, etc. introduces a large amount of pollutants to the atmosphere. These are carbon monoxide, carbon dioxide, nitrogen oxides, ammonia, hydrocarbons, chlorine, fluorine, lead, mercury, cadmium, particulate matters, etc. For example, oxides of sulphur and florine are released from fertilizer industries, chlorine, sulphur dioxide, nitrogen oxides, are released from cotton industries.

ii. Vehicular pollution

It is the main source of environmental pollution in big cities. Burning of dissel, petrol, etc. for running buses, cars, motor cycles, trains, etc. releases an immense amount of pollutants like carbon monoxide, carbon dioxide, nitrogen oxides and particulate matters (dust, dirt, etc.) into the air.

iii. Agricultural activities

Ammonia is a byproduct from agriculture-related activities. It is one of the most hazardous gases in the atmosphere. Use of fungicide (Sulphur), insecticides (Endosulfan), pesticides (DDT, BHC), etc. to protect our crops causes environmental pollution. Most of them are non-degradable pollutants which are not decomposed by micro-organisms. Farmers in Haryana and Punjab burn crop residue (stubble) to sow the next crop. It is an easy, fast and cost effective option to remove the stubble and to prepare soil bed for the next crop. But this stubble (paddy residue called '**parali**' in local language) burning has been cited as one of the major sources of air pollution in Delhi.

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iv. Mining operations

Mining activities cause massive air pollution by releasing respirable dust, harmful gases, (methane, oxides of nitrogen, etc.) and chemicals into the atmosphere. It adversely affects the health of mine workers and nearby residents.

v. Domestic Sources

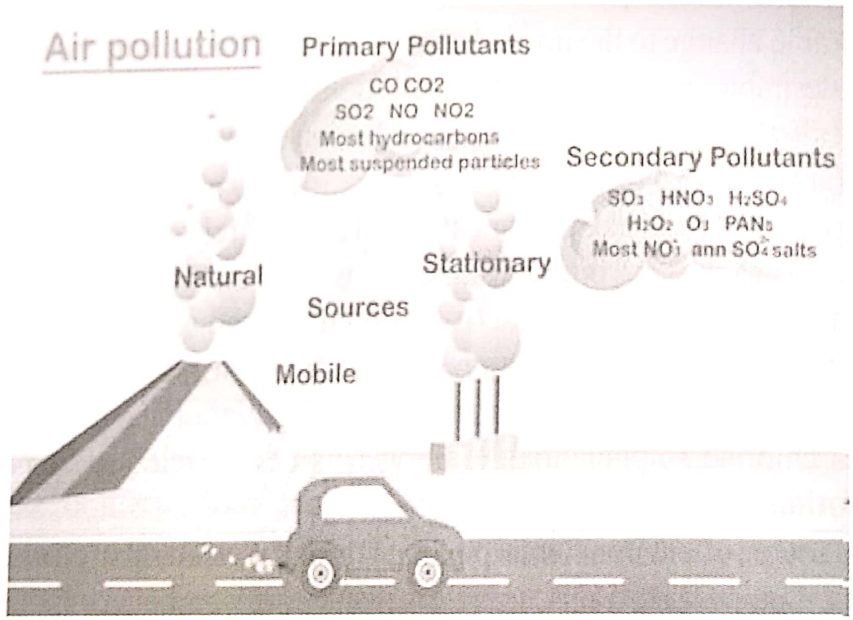
The sources of household air pollution include, the indoor burning of wood and cow-dung cakes for cooking and heating, seasonal open fires, the household cleaning chemicals, paints, etc. When the 'freon' (a coolant containing CFCs used in old refrigerators, air conditioners, etc.) is not properly disposed of, it leaks into the atmosphere and pollutes it. Aerosol spray-cans produced in many countries utilize CFCs as propellant which causes air pollution.

vi. Nuclear testing

Radioactive substances like Sr, U, Ra, etc., and the ionising radiations produced by atomic explosion and testing of atomic weapons contaminate the air.

vii. Airplane pollution

Aircraft engines release CO_2, flurocarbons, nitrogen oxides, etc. into the air. The 'halon', (a coolant containing halogenated hydrocarbons) used in aircraft engine causes destructive reaction with ozone .



(Fig.1.2 Air Pollution)

Effects of Air Pollution

Air pollution is the single largest environmental threat to human health. As a result of inhaling toxic air, our lungs become dustbins. If we keep polluting air, it will stop our breath for ever. Some of its effects are as follows :

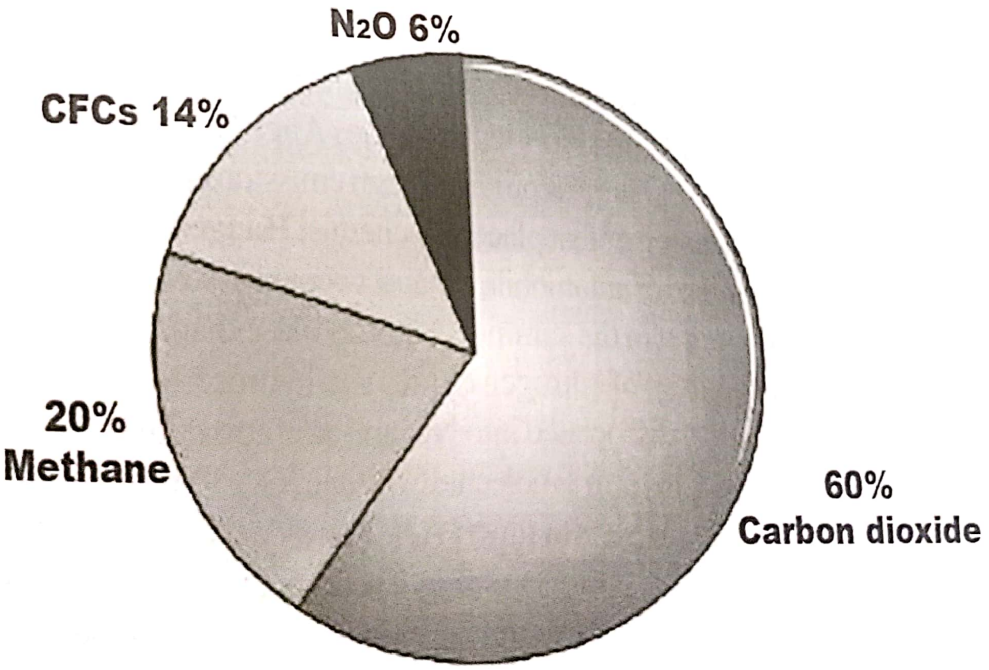
❖ Increase in concentration of CO_2 causes **green** house effect which results rise of temperature in the earth's atmosphere (global warming).

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❖ Increased concentration of CO decreases the oxygen carrying capacity of haemoglobin of the blood resulting oxygen deficiency in body tissues (hypoxia).
❖ SO_2, NO, NO_2, etc, released from industries come to earth as acidrain (HNO_2 and H_2SO_4) and affect flora, fauna and also cause damage to buildings and monuments.
❖ Lead (Pb) can damage the lungs of young children.
❖ Chlorofluro carbons (chemical compounds made up of chlorine, fluorine and carbon) cause destruction of ozone layer (ozone hole). So the ultraviolet rays reach in our body and cause skin cancer.
❖ Ionising radiations and radioactive pollutants cause cancer by mutations.
❖ Excess dust, smoke, CO_2 in the atmosphere and depletion of ozone layer affect our climate.
❖ Primary pollutants react to form secondary pollutants which are more potent than the primary pollutants, *e.g.*, photochemical smog.
❖ Sulphur dioxide (SO_2) and nitrogen oxides (NOx) can cause leaf injury, stomatal damage, premature senescence in plants, decrease photosynthetic activity (chlorosis), etc.
❖ Due to air pollution, many animals are forced to seek food in environments unknown to them.

Greenhouse Effect

Under normal condition, the surface of earth, atmosphere maintains the heat by radiating solar energy back into the space. However, when the concentration of carbon dioxide increases, it forms a thick cover and functions like glass panels of green house preventing the heat from being radiated back. It is the characteristic of carbon dioxide to trap long wave radiations. This heating effect in the earth;s atmosphere is called green house effect. An increase in temperature by 2-3 degree centigrade would cause melting of ice in poles and floods in coastal areas. Besides CO_2 also SO_2, NO,NO_2, CH_4, CFC, etc. contribute to green house effect.



(Fig.1.3 Relative contribution to Greenhouse effect)

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Acid Rain

Burning of fossil fuel (coal, oil, natural gas) in powerplants and by vehicles, and the smelting of metal ores are the major causes of acid rain. These human activities release oxides of nitrogen (NO, NO_2) and sulphur (SO_2, SO_3) into the atmosphere. Natural processes like volcanic eruptions, forest fires, lightening, etc. also add these pollutants to the atmosphere. They travel long distances in the earth's atmosphere and get oxidised into H_2SO_4 and HNO_3. These acids are dissolved in atmospheric water and fall back to the ground as acid rain or other forms of precipitation (such as snow, sleet, hail, and fog). On an average 60-70% of the acidity is ascribed to H_2SO_4 and 30 to 40% to HNO_3. It causes acidification of water bodies hence, affects aquatic life and human being. Acid rain destroyed fish populations in thousands of lakes in Canada and Sweden, leaving many of these as '**Fish graveyards**'.

Depletion of Ozone Layer

When the rate of destruction of ozone exceeds the rate of formation, it causes ozone layer depletion. The chloroflurocarbons (e.g. freons) are used as coolants in refrigeration and air conditioners. These are also used as solvents in industrial cleaning products, a blowing agents in fire extinguishers, propellants in aerosols, etc. The CFCs in the atmosphere are eventually broken down by UV rays from the sun, releasing free chlorine. Chlorine molecule acts on ozone and converts it into oxygen. One atom of chlorine can destroy upto one lakh molecule of ozone. About 80% of total stratospheric ozone depletion is due to emissions of CFCs only. The ozone layer plays a major role in climatology and biology of earth. Its depletion leads to temperature change and rainfall failure in the earth and consequent damage to life system. Ozone layer is the '**UV Blocker**'. It is the 'nature's sunscreen'. It protects us from harmful radiations of sun. Due to its depletion, the UV rays of sun reaches the Earth's surface and causes skin cancer, cataract, etc. The UV rays also affects the crops. In 1975, atmospheric scientists first discovered the ozone hole maximum over Antartctica.

Photochemical Smog

A new kind of air pollution called photochemical smog has emerged in the twentieth century, which is neither smoke nor fog. It was first reported in the Los Angeles, hence called **Los Angeles smog**. Increased use of vehicles in Los Angeles produces high emissions of NOx and hydrocarbons which is the principal cause of the smog of that place. Biochemist **Haagen-Smit** found that the Los Angeles smog was caused by reactions of automobile exhaust vapours in sunlight. As it is temperature dependant, the smog is most common in the summer. Automobile exhausts, together with power plants, are the most significant sources of nitrogen oxides and hydrocarbons of the atmosphere. When NO_2 absorbs visible light, it is dissociated into NO and atomic oxygen. The highly reactive atomic oxygen may combine with O_2 to form a molecule of ozone (O_3). With exposure to ultraviolet radiation, the NO_2 reacts with hydrocarbons to form PANs (peroxyacyl nitrates). NO_2 along with ozone and PAN appear in the form of an yellowish smog. Even small traces of the smog can affect the respiratory tract of humans and animals, and damage crops and trees. Besides Los Angeles, this smog is also common in cities like Sydney, Mexico, Beijing, etc. In 1952, London smog disaster is said to have caused death of 4000 people.

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Control Measures of Air Pollution

We pollute the air and burn up our future. Some control measures are:

1. Emphasis on clean energy resources

More use of clean energy sources like solar, wind and geothermal energy are the best ways to curb air pollution. Governments of various countries have been providing grants to consumers for this purpose.

2. Selection of fuel

Coal can be substituted by oil to prevent formation of smoke.It is wiser to replace coal and petrol by solar energy and wind power. Vehicular pollution can be minimised by using new proportion of gasoline and air, unleaded petrol,using gas additives to improve combustion.

3. Use of energy efficient devices

LED lights consume less electricity as against their counterparts. Use of these lights helps in reducing pollution by consuming less energy.

4. Plantation

Citizens of the country should be encouraged to plant more trees. Plants reduce carbon dioxide and a large number of other pollutants in the air. Planting trees in areas of high pollution is an effective measure to curb air pollution. *When plants are grown with a lot of care, humans can breathe fresh air .*

5. Use of public mode of transportation and restriction on private vehicles

Using public transportation as much as possible will reduce pollution.The **Odd-Even Policy** on private vehicles was first introduced in 2016 by the Govt. of Delhi. As per the rule, between 8am and 8pm private, non-commercial vehicles are allowed out on the roads on alternate days – odd days if the last digit of the vehicle's registration number is an odd number, and even days if it ends in an even number.

6. Understanding the concept of Reduce, Reuse and Recycle

All used items should not be thrown away, because most of them can be used for some other purpose. *e.g.,* old jars can be used to store pulses, cereals, etc.

7. Use of equipments

Use of electrostatic precipitators, inertia separaters, filters, scrubbers, etc., can separate the pollutants at the source. Catalytic converters convert CO and NO_2 to less hazardous pollutants.

i. **Electrostatic precipitator**

It is an electrical device to remove particulate matter present in the exhaust of thermal power plants. More than 99% particular matter can be removed by using it.It has electrode wires and a stage of collecting plates. Electrode wires are provided with an electric current of several thousand volts to releases electrons.These electrons attach to dust particles and give them a negative charge within a very short time. Collecting plates are earthed so that they attract charged dust particles.When the air passes through collecting plates, the dust particles fall on it.

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ii. **Scrubber**

It is an air pollution control devices that use liquid to remove particulate matter or gases li sulphur dioxide from industrial exhausts. The contaminated exhaust is passed through a spray water and the water absorbs the contaminates. Different types of liquids can be used to effectiv remove the target contaminates.

iii. **Catalytic converters**

These are air pollution control devices fitted into automobiles for reducing emission of tox gases like NO_2 and CO. They have metals like platinum, palladium and rhodium as catalyst. As th exhaust emission passes through a catalytic converter, it tackles nitrogen oxide and splits into nitrog and oxygen (harmless) due to reduction. The other catalyst oxidises carbon monoxide into carb dioxide. Automobiles with catalytic converters require to use unleaded fuel, because the lea prevents it from taking up the pollutants.

Other measures

- The eco-friendly industrialization should be encouraged.
- Population explosion should be checked.
- Pollution control laws should be strictly enforced.
- Nuclear explosion should be banned.
- Burning leaves, trash, and other materials should be avoided.

The air pollution is damaging our planet. We should realize its effects and prevent this damag

Some International Initiatives

i. Montreal Protocol

It was a global environmental agreement that regulates the production and use of nearly 10 man-made chemicals that contribute to the depletion of Earth's ozone layer. It provided a set o practical, actionable tasks to protect stratosphere ozone that were universally agreed on. It wa first adopted in September 1987.

ii. Rio de Janeiro Earth summit

The United Nations Conference on Environment and Development (UNCED) , also know as the; 'Rio de Janeiro Earth Summit', was held in Rio de Janeiro, Brazil, from 3-14 June 1992 The 'Earth Summit' concluded that the concept of sustainable development was an attainable goa for all the people of the world. An agreement was made "not to carry out any activities on the land of indigenous peoples that would cause environmental degradation or that would be culturally inappropriate".

iii. Kyoto protocol

It was first signed on, 11th December 1997, Kyoto, Japan. The targets for the first commitmen period to reduce the onset of global warming by reducing emissions of the six main greenhouse gases, namely carbon dioxide (CO_2), Methane (CH_4), nitrous oxide (N_2O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), and sulfur hexafluoride (SF_6). The aim is to prevent dangerous anthropogenic interference with the climate system.

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Water Pollution: *Makes the Nature Sick*

Our earth is the only planet to have liquid water on its surface. According to studies, approximately 71% of the entire earth is covered with water. It is the most important resource for survival on a planet.It is the most important and abundant constituent of protoplasm of living cells which allows them to function properly. So the life cannot exist without water. Any change in physical, chemical and biological quality of water of different bodies so that it becomes unsuitable for organisms is called water pollution.

Sources of Water Pollution

Water pollution has many sources. some important sources are:

i. Industrial effluents ii. Municipal sewage iii. Agricultural particles
iv. Inorganic fertilizers v. Ground water pollution vi. Costal water pollutions

i. Industrial effluents: Waste generated by manufacturing or chemical processes in industries is a huge source of water pollution. Industrial waste disposal has been a major problem since the industrial revolution. In fact, a number of large- and medium-sized industries do not have enough waste treatment facilities. The industrial effluents carrying both organic and inorganic hazardous substances, enter the water bodies and contaminate them. The pollutants from industrial sources include asbestos, lead, mercury, nitrates, phosphates, sulphur, oils, etc.

ii. Sewage: Waste water that frequently contains faeces, urine and laundry waste is called sewage. Its disposal is a major challenge in the developing countries. Untreated sewage water contains organic wastes, suspended particles, liquid pollutants, harmful viruses and bacteria, etc. which contaminate the water bodies like rivers, ponds and seas.

iii. Agricultural particles: The agriculture is both the source and victim of water pollution. Pesticides, insecticides, animal excreta, etc., used for agricultural purpose contaminate the water. The use of wastewater and polluted surface water contaminates the crops and transmit disease to humans.

iv. Inorganic fertilizers: Fertilizers used in the crop fields contain inorganic heavy metals which are accumulated in the food chain. The increased use of fertilizers also cause eutrophication of stagnant water bodies.

v. Ground water pollution: The groundwater is normally clear and clean as the particulate matters are filtered by the ground. But the industrial discharges, leaking fuel tanks or toxic chemical spills, pesticides and fertilizers applied in agricultural fields migrate to the water table and contaminate groundwater.

vi. Costal water pollution: Discharge of rubbish from ship, leakage of oil tankers etc pollute the ocean or coastal waters. Most of the pollutants of the sea come from the land. Waste water from industries, oil drops from motor vehicle engines make their way to the sea and and contaminate the sea water.

Effects of Water Pollution

Clean water is the base of our life.The water we pollute will find its way back to us.If we continue polluting water, then we have to pay a big cost in future. Some of its effects are :

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It has been observed that concentration of DDT increases gradually and its accumulation is highest at the top trophic level. Similarly concentration of Hg increases, when it passes through the food chain. One part of mercury in the phytoplanktons gets magnified to about 34'600 parts in the fishes. Accumulation of toxic substances through biomagnification makes humans body more sensitive to cancer and also creates kidney, liver , heart and lungs disorders. Minamata disease in Japan (1953) is neurological disease caused due to biomagnification of Hg.

Minamata disease

Minamata disease was first discovered in Minamata city, Japan in 1956. It is an unusual neurological disorder caused due to mercury poisoning. The symptoms of the disease includes ataxia, numbness in limbs, muscle weakness, etc.. In extreme cases it leads to paralysis, coma and death. The Chisso Corporation, a Japanese chemical company, opened a chemical factory at Minamata city which started acetaldehyde production in 1932. The company used mercury as a catalyst. The industrial wastes containing mercury were dumped into Minamata Bay. The toxic non-degradable mercury present in the industrial waste entered the food chain. Its concentration was increased in fishes due to bio-magnification (phytop lanktons ⟶ Zooplanktons ⟶ Small fishes ⟶ Large fishes). Local people were affected with neurological problems like numbness of limbs and damage of brain by eating shellfish and fish from the Minamata bay. The neurological syndrome caused by extreme mercury poisoning is now known as Minamata disease. Hundreds of people died, and thousands were affected.It also caused death of cats, dogs, crows, etc., who ate scraps from the family table.

Control Measures of Water Pollution

It can be controlled by :

(i) Sewage treatment (ii) Treatment of industrial waste (iii) Recycling of waste

(iv) Removal of pollutants(iv) Composting (v) Use of Aquatic plants (vii) Legislation

1. Sewage treatment: It is the process of removing contaminants (suspended particulates, toxic chemicals, nitrates, phosphates pathogenic bacteria, etc.) from the sewage (municipal waste water). This treatment involves physical, chemical, and biological processes.

i. Primary treatment: It is a physical process by which the large suspended particles are removed. It involves two processes, i.e, sedimentation and filtration. The wastewater is placed in a holding tank, solids are settled at the bottom due to gravity and are separated.

ii. Secondary treatment: It removes small soluble or suspended particles from the waste water. The organic impurities are consumed by aerobic bacteria producing carbon dioxide, water, and energy. Air is circulated to stimulate bacterial decomposition.

iii. Tertiary treatment: It is designed to filter out substances that might damage sensitive ecosystems. Phosphates, ammonia, nitrates, etc., are removed by this treatment.

2. Treatment of industrial waste: Different processes like neutralization, removal of toxic compounds, coagulation of impurities, precipitation of metals, reduction of temperature, etc. are followed for treatment of industrial waste.

3. Recycling and reutilization of waste water: Waste water may be reused for different purposes like irrigation, groundwater recharge, recreational purposes, etc.

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1. Accumulation of sewage causes growth of micro-organisims, hence, spreads epidemic diseases like cholera, diarrhoea, etc. So if we pollute water, it will put us on the bed.
2. Accumulation of organic sewage and fertilizers causes **eutrophication** of water bodies.
3. Drinking water rich in nitrites causes methamoglobinaemia.
4. Insecticides, pesticides (DDT, BHC, etc.) and other pollutants (e.g., Hg) that cannot be degraded undergo biomagnification, affecting human health and other desirable species.
5. Oxides of nitrogen and sulphur released due to burning of fossil fuel causes acid rain which reduces soil fertility,causes acidification of water bodies and damage to building, monuments etc.
6. Poisonous chemicals kill aquatic organisms responsible for bio-purification of water. Hence, the self purifying ability of water is lost.
7. Mercury (Hg) is a non-biodegradable pollutant which affects nervous system and causes damage to brain (*e.g.,* **Minamata disease** in Japan).
8. Use of drinking water rich in fluoride causes fluorosis.
9. Exposure to high levels of lead may cause anemia, kidney and brain damage. It can damage the nervous system of the foetus, if the pregnant woman is exposed to lead.
10. Chronic exposure to cadmium affects kidney, bones and lungs. The disease **Itai-Itai** characterised by low grade of bone mineralization (osteoporosis) and intense bone associated pain was due to excess exposure to cadmium.

Eutrophication

Eutrophication(*Greek eutrophos, "well-nourished"*) is a condition where stagnant water bodies become overly enriched with minerals (particularly phosphorus and nitrogen) and nutrients, usually caused by runoff of animal wastes, fertilizers, sewage etc. from the land. The rain acts as an agent by washing these leachable nutrients from the soil into water bodies. It induces excessive growth of algae (**algal bloom**). After their death, the dead organic matter becomes food for micro-organisms (*e.g.,* bacteria, fungi, etc.) that decompose it. The number of zooplankton, bacteria, fungi increases significantly. They use up more oxygen dissolved in the water. It causes deficiency of dissolved oxygen in water bodies and adversely affects fishes, plants and other aquatic animal species. Growth of toxic algae also causes potential damage to the animals and human using this affected water for drinking and bathing purpose. It can be prevented by preventing the introduction of nutrients and limiting the phosphorus loads (limiting factor for the growth of algae) of these stagnant water bodies.

Biomagnification

Biomagnification is the process of increase in concentration of non-degradable pollutants (*e.g.* mercury, copper, lead, cadmium, DDT etc.), when passing through successive trophic levels in the food chain. For example, We use DDT to kill pests in the crop fields. When rain water flows over this field, the DDT enters the pond ecosystem. Then it is absorbed by producers and its concentration increases when it passes through trophic levels of the food chain.

(Phytoplankton ⟶ Zooplankton ⟶ Small fishes ⟶ Large fishes)

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4. Removal of pollutants: Different processes like absorption, electrodialysis, ion exchange, reverse osmosis help in the removal of pollutants from waste water.

5. Composting: The organic waste is biodegraded by microbes and is converted into compost. It is one of the best management practices of reducing runoff and improving downstream water quality.

6. Use of aquatic plants: Some aquatic plants(e.g, water hyacinth) are very efficient in purifying domestic and industrial waste water. They absorb soluble heavy metal ions and cyanides from polluted water and help in purifying the water bodies.

7. Legislation: The Water (Prevention and Control of Water Pollution) Act should be strictly enforced. *Water is indispensable for life. We shouldn't let our future dry up by polluting this precious gift of nature.*

Primary sewage treatment	Secondary sewage treatment
1.It is a purification process involving physical process.	1.It is a purification process involving the microbes.
2.It removes large suspended particles.	2.It removes small dissolved or suspended particles.
3. It occurs by sedimentation and filtration.	3. Biological breakdown occurs by microbes.
4. It is inexpensive and relatively less complicated.	4.It is very expensive and complicated.
5. It takes less time to complete the process.	5. It takes more time to complete the process.

Biological Oxygen Demand(BOD)

- The amount of dissolved oxygen consumed by microorganisms for decomposing organic matter in aerobic condition.
- The more organic matter (pollutants), the greater the BOD and the greater the BOD, the lower the amount of dissolved oxygen available for higher aquatic animals .
- It is a reliable gauge of the organic pollution of a body of water.

Chemical Oxygen Demand(COD)

- Amount of oxygen required for the chemical oxidation of total organic matter in water.
- It is a chemical process.
- It is greater than BOD.
- Complete degradation of pollutants occurs.

BOD	COD
1. Amount of O_2 consumed by bacteria for decomposition of organic matter in aerobic condition.	1. Amount of O_2 required for the chemical oxidation of total organic matter in water.
2. It is a biological process.	2. It is a chemical process.
3. Its value is lower than COD.	3. Its value is more than COD.
4. Incomplete degradation of pollutants.	4. Complete degradation of pollutants occurs.

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